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Human stress classification during public speaking using physiological signals

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ABSTRACT

Public speaking is a common type of social evaluative situation and a significant amount of the population feel uneasy with it. It is of utmost importance to detect public speaking stress so that appropriate action can be taken to minimize its impacts on human health. In this study, a multimodal human stress classification scheme in response to real-life public speaking activity is proposed. Electroencephalography (EEG), galvanic skin response (GSR), and photoplethysmography (PPG) signals of forty participants are acquired in rest-state and during public speaking activity to divide data into a stressed and non-stressed group. Frequency domain features from EEG and time-domain features from GSR and PPG signals are extracted. The selected set of features from all modalities are fused to classify the stress into two classes. Classification is performed via a leave-one-out cross-validation scheme by using five different classifiers. The highest accuracy of 96.25% is achieved using a support vector machine classifier with radial basis function. Statistical analysis is performed to examine the significance of EEG, GSR, and PPG signals between the two phases of the experiment. Statistical significance is found in certain EEG frequency bands as well as GSR and PPG data recorded before and after public speaking supporting the fact that brain activity, skin conductance, and blood volumetric flow are credible measures of human stress during public speaking activity.

1. Introduction

A wide range of stress-related diseases have impacted the life of a large segment of the population and is significantly increasing every year. Human stress has damaging effects on the mental and physical health of individuals [1,2]. Human stress tends to increase the risk of many diseases such as heart attacks, diabetes, strokes, sleep disorder, and can even result in causing sudden death [3–6]. A study conducted by World Health Organization (WHO) presented the fact that medical and social effects associated with stress are increasing at an alarming rate and are causing an impact not only in the lives of adults but also in young individuals and kids [7]. Therefore, the development of reliable and efficient stress assessment methods is an exciting research topic in the field of physiology, neurology, and biomedical engineering nowadays [8].

Human stress can be classified into chronic and acute stress [9]. Chronic stress is instigated due to continuously facing a stressful situation for over a period of months e.g., job strain, challenging relationships, and bad financial conditions [10]. On the other hand, acute stress is an instantaneous response of the human body triggered by a stressful situation e.g., public speaking, mental arithmetic task, and interview or viva [11–13]. When a person faces an acute stressor, the human body responds by providing the required physiological and immune system adaption to cope with the consequences to achieve the new mental equilibrium state required for survival [9]. Changes in the autonomous nervous system of the human body take place in response to stressors thus triggering diverse body changes, which have been measured by different kinds of stress markers to date [14].

Stress is an inexorable phenomenon and a prolonged exposure to stress can severely impair human health. Among the various factors causing human stress, stress instigated by social evaluative situations like an interview or a public speaking activity significantly affects the confidence and well-being of an individual. Therefore, it is important to develop a robust strategy to curb and manage human stress to help individuals to gain lost mental and physical equilibrium. According to the National Institute of Mental Health, around 73% of the population is affected by fear of public speaking and it has a significant impact on the fulfillment of the duties throughout their lifespan. Even though only

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very few individuals enjoy public speaking, but people having social anxiety disorder find these social evaluations quite unbearable. Such individuals perceive that they are always going to get negative feedback from the person before them and due to these negative expectations they strive to avoid these performance assessment situations at any cost. Evading these kinds of situations neither does any good to a person's well-being nor helps in coping with such judgemental situations in the future. Running away or deliberately avoiding such situations is not the solution to manage psychosocial stress, rather a better course of action to mitigate the effects of this stress needs to be formulated.

A wide range of stress measurement markers is being used by the research community as well as medical practitioners. Psychometric questionnaires are considered as a suitable method by the physiatrists, psychologists, and physicians for the assessment of stress [15]. Commonly used stress measurement questionnaires include perceived stress scale (PSS) [16], state-trait anxiety inventory (STAI) [17], relative stress scale (RSS) [18], daily stress inventory (DSI) [19], brief symptom inventory [20], and trier inventory for the assessment of chronic stress (TICS) [21]. On the other hand, the biomedical research community believes that chemical bio-markers like α-amylase and copeptin are more reliable for stress assessment [22,23]. Even though psychometric questionnaires and biochemical markers are considered reliable stress measurement tools for specific applications but they have some shortcomings like continuous stress monitoring and self-perception of the patient about a particular condition. Stress can also be measured using a non-wearable sensor like Microsoft kinect [24]. Another most commonly used stress assessment method is by measuring the physiological changes in the human body occurring as a result of a stressor with the help of wearable sensors. Stress measurement using physiological signals has been performed in the literature using electrocardiography (ECG) [25], electroencephalography (EEG) [26], photoplethysmography (PPG) [27], galvanic skin response (GSR) [28], skin temperature (ST) [29], heart rate (HR) [30], heart rate variability (HRV) [31], blood volume pressure (BVP) [32], electromyogrpahy [33] and by fusion of these bio-signals [34].

In literature, several acute human stress measurement studies using different stimuli have been proposed. The framework used for the assessment of human stress in the literature follows the following steps. Firstly, physiological signals are acquired by inducing stress either in laboratory settings [35–38] or in real life situations [39–43]. Followed by this, a discriminating set of features are extracted to classify human stress by either using some pattern recognition or machine learning techniques. Subsequently, a diverse range of machine learning techniques has been used to classify human stress into two and three levels. The public speaking activity has been used as a stress-inducing stimulus in various stress measurement studies [44,45]. An acute stress measurement scheme using unobtrusive physiological markers of skin temperature, heart rate, and pulse wave in response to the mental arithmetic task and public speaking stimuli was presented in Ref. [11]. The study concluded that the rest state (27.86 auxiliary units) can be discriminated from stress condition (47.55 auxiliary units) using these unobtrusive physiological bio-markers. Another human stress measurement study using HRV signals was presented in Ref. [46] and data was annotated using scores of the STAI questionnaire. A human stress assessment scheme using GSR and PPG signals in response to stimuli of cognitive task and public speaking was presented in Ref. [47]. Human stress assessment using physiological signals (GSR, PPG, and HRV) and sociometric sensors (microphone, accelerometer, proximity, and infrared sensors) in response to a cognitive task and public speaking stimuli were presented in Ref. [48]. Another study for analyzing the real-life public speaking anxiety using recurrence quantification analysis of the GSR signals was presented in Ref. [49]. Non-linear features of the GSR signals and neural network were used to identify the calm and the anxious state as well as low anxiety and high anxiety state. Temporal parametric models to quantify the bio-behavioral trajectories of public speaking anxiety using real-life public speaking stimuli and physiological signals of GSR, PPG, and HR was presented in Ref. [50]. The study concluded that the proposed time-based analysis of the bio-behavioral measures was more robust in estimating the characteristics of the participants as compared to the traditional model.

The multimodal human stress assessment methods have remarkable results in terms of obtained accuracy as compared to single modality-based methods [51,52]. Multimodal stress assessment schemes in response to different stimuli have been proposed in a wide range of studies available in the literature. Human stress measurement using a fusion of near-infrared spectroscopy (NIRS) and physiological signals of PPG, ECG, and seismocardiogram in response to mental arithmetic and memory tasks was presented in Ref. [53]. An accuracy of 85% was achieved for three states using a random forest (RF) classifier. Assessment of mental stress using a fusion of EEG and functional-NIRS signals in response to mental arithmetic task with an accuracy of 95.1% using SVM classifier was proposed in Ref. [54]. The fusion of EEG signals with other psychological modalities like ECG [55] and galvanic skin response [56] has been used for boosting the performance of stress classification. The fusion of GSR and ST has been used to classify human stress with an accuracy of 84% [57]. Heart rate, skin conductance, and accelerometer sensors have been adopted for the measurement of real-time stress recognition using daily-life activities [58]. An accuracy of 92.15% was achieved for three-level stress classification using the fusion of features from these three modalities. A novel multimodal stress recognition scheme using a fusion of features from BVP, HR, ST, GSR, and respiratory rate (RR) in response to the mental arithmetic task has been proposed in Ref. [59]. An accuracy of 83% and 72% is achieved for two and three class stress classification, respectively. Some of the other stress measurement studies available in the literature include fusion of ECG and contextual data obtained from smartphones [60], a fusion of accelerometer and electrodermal activity (EDA) data [61], a fusion of respiratory, ECG and accelerometer sensors data [62] and fusion of accelerometer, EDA and Bluetooth sensors data [63].

Acute stress classification schemes available in the literature have used a wide variety of stressors (Stroop color-word test [64], trier social stressor test [11], mental arithmetic tasks [54], public speaking [65], music [66] and Montreal imaging stress task [67]) and physiological sensors (GSR, ECG, PPG, EEG, ST, HR, and HRV) to provide useful insights about stress assessment. Physiological signals have been used both in the individual and fused manner in a significant number of stress detection studies conducted in response to a public speaking stimulus. However, an association of EEG and fusion of EEG signal with other physiological markers against public speaking stimuli is still under-examined and has been a focus of only very few studies even though EEG signal has been found to an effective marker of human stress. The existing schemes investigating the alliance between physiological signals and public speaking have discrepancies in their finding when they are compared with the response recorded via subjective questionnaires. Moreover, systematic limitations of the existing schemes like small sample size, computational complexity, high cost of the data acquisition equipment are some salient factors, which need to be taken care of. To the best of our knowledge, no acute stress classification study using a fusion of data obtained from EEG, GSR, and PPG sensors in response to real-life public speaking activity has yet been performed. To this end, we propose an acute stress classification framework using a fusion of features extracted from physiological signals of EEG, GSR, and PPG in response to a real-life public speaking activity. EEG, GSR, and PPG data of the subjects are recorded both in the rest state and during public speaking activity. Frequency domain features are extracted from the EEG data, whereas time domain features are extracted from the GSR and PPG signals. Features extracted from all the physiological modalities are fused to develop a more accurate and robust human stress classification framework. Five different classification algorithms, which include k-nearest neighbors (kNN), decision tree (DT), random forest (RF), multilayer perceptron (MLP), and support vector machine (SVM) are employed to classify human stress. To get the better of the mentioned

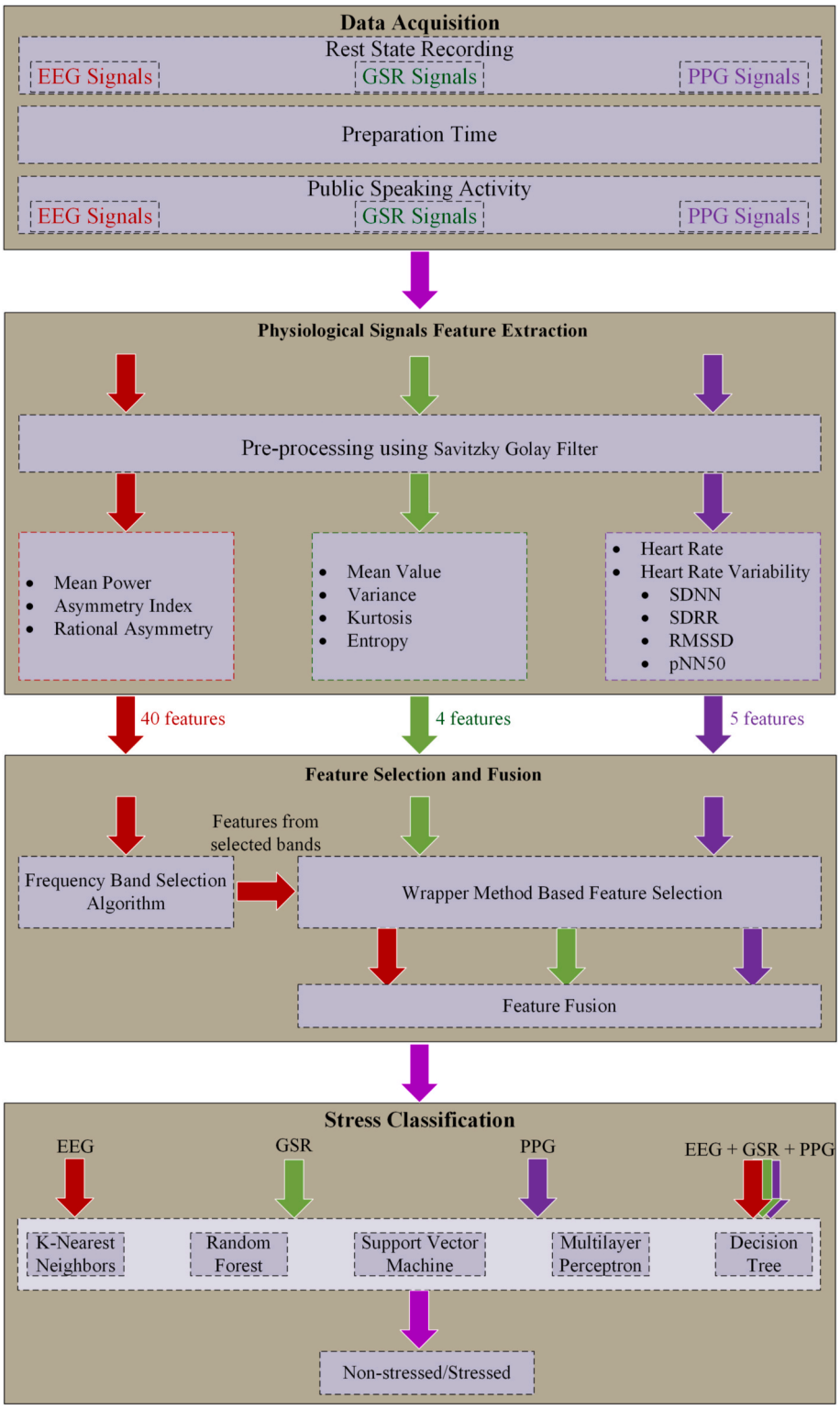

Fig. 1. Block diagram of the proposed multimodal stress detection scheme.

challenges, acquisition of the physiological signals is performed using a low-cost commercially available MUSE EEG headband, and Shimmer GSR and PPG optical pulse clip. Moreover, computational complexity is attended by appropriately applying band and feature selection to the acquired physiological signals. To conquer the effect of low sample size in the existing methods, physiological signals are acquired from a decent-sized population of forty subjects, which is substantially large than the sample size of the latest stress classification schemes.

The rest of the paper is organized in the following sequence. The proposed human stress detection framework is presented in Section 2, which includes details about the data acquisition protocol, feature extraction, feature selection and fusion algorithms used, and details about the classifiers adapted for stress classification. Section 3 presents the result and analysis of the proposed scheme followed by a comparison of the proposed scheme with other methods. Finally, Section 4 presents the conclusion and future work of the proposed scheme.

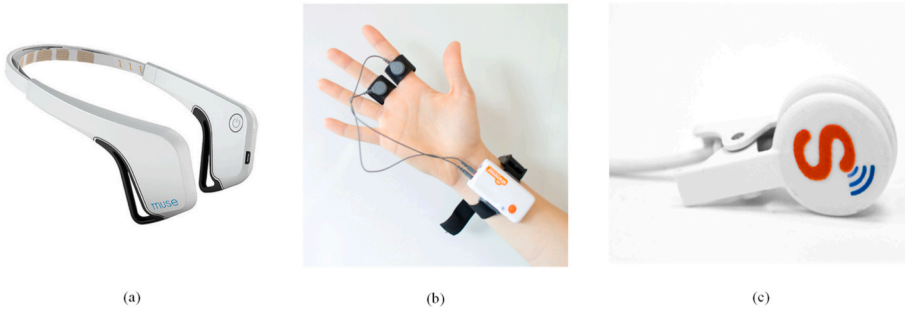

Fig. 2. Physiological sensors used for data acquisition (a) MUSE EEG headband (b) Shimmer GSR module (c) Shimmer PPG ear clip.

2. The proposed framework

Fig. 1 shows the block diagram of the proposed framework that includes data acquisition, feature extraction, feature selection and fusion, and stress classification. The following subsections elaborate on the details of each block of the framework.

2.1. Data acquisition

Physiological signals were recorded from forty mentally and physically healthy subjects consisting of twenty males and twenty females in two different states i.e., rest-state and while performing public speaking activity. The participants of the study were either university students or instructors having an age ranging from 18 to 40 years. The mean and standard deviation of participant's ages are 24.85 and 6.69 respectively. All the participants were literate and had at least attended college classes. The participation of the subject in the study was completely voluntary and they were allowed to leave the experiment at any stage.

In this study, we have used data acquisition equipment designed by two different vendors i.e., MUSE (version 2016) designed by Interaxon Incorporated, and GSR + module and PPG optical pulse clip developed by Shimmer Research Ltd. MUSE is a wireless, lightweight, easy to wear, flexible and portable EEG headband widely used both for meditation and research purpose. MUSE headband is composed of four electrodes, two of which are frontal (*AF*7 and *AF*8) and two are temporal (*TP*9 and *TP*10) in addition to the reference electrode at location *Fpz*. MUSE headband sends data by using Bluetooth technology at a sampling rate of 256 Hz via Muse monitor (an android application downloadable from https://mind-monitor.com). Shimmer GSR + module is used to acquire the data from the GSR and PPG sensors at a sampling rate of 256 Hz. GSR sensors are placed on the fingers and PPG optical pulse clip is placed on the ear lobe of the subjects. GSR and PPG data is acquired by using the ShimmerCapture desktop application via a Bluetooth connection downloadable at https://www.shimmersensing.com/products/shimmercapture. Fig. 2 shows the equipment used for data acquisition in this study.

The experimental protocol starts with a briefing about the experiment to the subjects and getting their consent as well as demographic details in a written manner. After the briefing, the participant was asked to be seated comfortably so that their baseline physiological data can be acquired. Before data acquisition, the EEG headband was placed on the user's forehead, the GSR sensor was placed on the fingers of the left hand and PPG optical pulse clip was placed on the ear lobe of the subject. The participant seated on a comfortable chair was asked to focus on a blank computer screen placed in front of them and the physiological signals were recorded for a duration of 3 min in the open eye condition. Next, the participant was given 10 min to prepare a talk which has to be presented for a minimum duration of 5 min in front of a real audience. After the preparation, the participant was escorted to another room for a presentation in which a sizable audience was present to listen to the talk. EEG, GSR, and PPG signals were acquired while performing the public speaking activity. The experimental protocol used in the study was designed according to the declaration of Helsinki and was certified by the Directorate of Advanced Studies, Research, and Technological Development at the University of Engineering and Technology, Taxila, Pakistan.

2.2. Physiological signals feature extraction

Before the extraction of features, the recorded physiological signals are preprocessed to remove the artifacts. The recorded physiological signals are subjected to artifact removal techniques to remove muscular artifacts, power line noise, and other artifacts. MUSE headband has an onboard noise cancellation scheme that classifies the data into clean or noisy signals based on statistical properties i.e., variance, amplitude, and kurtosis values of the acquired EEG data. If the statistical properties of the acquired EEG data have a value less than the predefined threshold then the signal is regarded as a clean EEG signal otherwise the signal is found to be noisy and discarded. Savitzky-Golay (SG) filter was applied to the recorded physiological signals (EEG, GSR, and PPG) for smoothing the data along with the preservation of signal strength. SG filter is based on the principle of least square polynomial approximation and is an effective method for data smoothing [68]. The pre-processed EEG data was transformed to different frequency (delta (0–4 Hz), theta (4–7 Hz), alpha (8–12 Hz), beta (12–30 Hz), and gamma (30–50 Hz)) bands using MUSE on-board signal processing module.

After pre-processing, features are extracted from the recorded physiological sensors data to classify human stress. Frequency domain features extracted from the EEG signals include mean power (MP), asymmetry index (AI), and rational asymmetry (RASM). Mean power is calculated by computing the average of the absolute power of each band for each channel of the MUSE headband. The total number of features obtained from MP is 20 i.e., five for each channel of the MUSE headband. AI is the difference in the absolute power of each band for the asymmetric channels of the left and right hemispheres, whereas RASM is the ratio of the absolute power of each band for the asymmetric channels of the left and right hemispheres. Each of the AI and RASM feature groups results in 10 feature values, from the EEG data. The total number of extracted EEG feature values is 40. For the GSR signal, four time-domain features are extracted from the recorded data which include mean (M), variance (V), kurtosis (K), and entropy (E) resulting in 4 feature values. From PPG data, heart rate (HR) and four different time-domain features that include the standard deviation of normal-to-normal intervals (SDNN), standard deviation of the time between two subsequent R-peaks (SDRR), root mean square of successive RR interval differences (RMSSD) and percentage of successive RR intervals that differ by more than 50 ms (pNN50). The feature values obtained from PPG data are of length 5 i.e., 1 value for heart rate and 4 values for HRV. A feature vector length (FVL) of 49 is obtained from all three modalities.

2.3. Feature selection and fusion

The extracted features from the EEG signals are fed to the band selection algorithm proposed in Ref. [8] to select the features from the

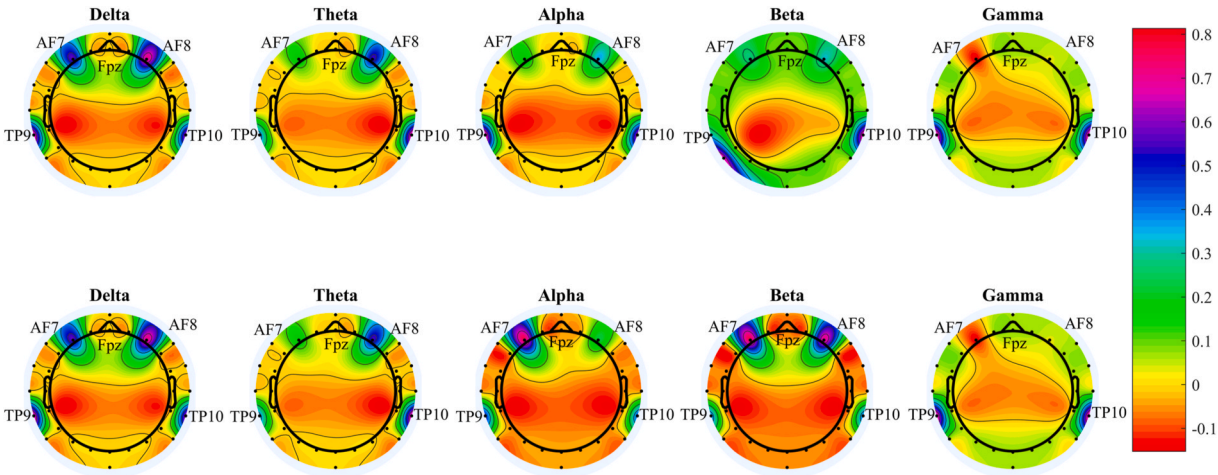

Fig. 3. Brain activity visualization in terms of power spectral density of different EEG bands for the two phases of experiment Row 1: rest-state, Row 2: during public speaking activity.

most appropriate frequency band. The band selection algorithm work by selecting features from the frequency band that maximizes the achieved classification accuracy from all the frequency band combinations. 1000 iterations are performed for each frequency band combinations to select the optimum band. The extracted features from the EEG beta-band are found to be the yielding the highest stress classification accuracy among all EEG frequency bands.

Wrapper method for feature selection with recursive feature elimination technique is then applied on the features from the beta band of the EEG signals and the features from the GSR and PPG data separately to select the optimum subset of features from each modality. Recursive feature elimination is a greedy feature optimization to select the highest performance feature subset. This technique built the feature selection model repeatedly in a recursive manner and keep track of the best and the worst performing features for each iteration. In the next iteration, the models are built by using only the left subset of features and this process continues till the point when all the features have been tested. Finally, the features are ranked according to their order of elimination to select the most discriminating subset of features. The selected subset of features from EEG data consists of MP of the beta band from AF7, AF8, and TP9 channels, AI of the beta band between AF7, and AF8 channel, and RASM of the beta band between AF7 and AF8 channels, which results in a total of five selected features from EEG data. Variance and entropy features are selected from GSR data, whereas for PPG data HR, SDNN, and RMSSD features are selected.

The selected subset of features from all modalities is fused to classify acute human stress. In this study, we have employed a late feature fusion (LFF) scheme in which each modality is applied upon with a feature selection algorithm and then the selected features from each modality are fused to form the final feature vector. The total number of selected features from EEG, GSR, and PPG are 5, 2, and 3 respectively which results in a FVL of 10 from all three modalities.

2.4. Stress classification

The feature vector obtained after feature selection and fusion is subjected to classification via different machine learning algorithms. SVM is a supervised machine learning algorithm, which classifies the data points by building an N-dimensional hyperplane between the two classes. Kernels for the SVM classifier used for classification in this study is radial basis function (RBF). k-nearest neighbors (kNN), decision tree (DT), random forest (RF), and multilayer perceptron (MLP) classifiers have also been used for the evaluation of the proposed stress classification scheme. kNN is a non-parametric-based supervised machine learning algorithm. kNN is termed as a lazy learning algorithm because it stores all the training examples and predicts the output based on some distance metric (euclidian distance in this study) of the test data point from the training examples. kNN with 5 nearest neighbors is used in this study. DT is another powerful and efficient supervised machine learning algorithm based on the entropy measure which classifies the data by building a tree composed of a root node, decision node, and leaf nodes. The topmost node of the tree is called the root node and leaf nodes of the data correspond to the label of the class or final decision, whereas decision nodes have further nodes which eventually lead to the leaf node. RF classifier is an extension of the decision tree and is an ensemble classifier in which there are multiple decision trees (hundreds of them) and the decision obtained from the majority of the trees in the forest is concluded to be the final decision. RF with 100 trees is used in the current stress classification scheme. MLP is the simplest type of feed-forward artificial neural network composed of an input layer, hidden layer, and output layer. An activation function is implemented in the hidden layer which transforms the input to produce the output. In this study, an MLP network with 4 hidden layers and a momentum and learning rate of 0.3 and 0.4, respectively is used.

3. Experimental results

In this section, we report the results of statistical analysis as well as stress classification using the fusion of selected features from three different modalities.

3.1. Statistical analysis

Statistical analysis of the physiological signals (EEG, GSR, and PPG) is performed by applying the paired *t*-test on the absolute power for each frequency band of the EEG signal obtained from all the channel of the MUSE headband, skin resistance obtained from the GSR sensor, and measure of blood flow obtained from PPG sensor during rest state and while performing public speaking activity. A null hypothesis and p-value are used to determine the *t*-test decision. The difference between the groups is considered significantly different if the p-value is less than 0.05 and the null hypothesis is 1. T-test applied on EEG signals reveal that only alpha and beta bands of the frontal channel *AF*7 (a p-value of 0.002 and 0.04, respectively), and beta band of channel *AF*8 (a p-value of 0.003) are found to be statistically significant. Fig. 3 shows the brain activity visualization obtained using the power spectral density of the EEG data of the participant during the two phases of the experiment. The red color in the brain maps depicts a strong cortical activation whereas the orange color depicts minimal brain activity. It can be observed from the brain maps that the alpha and beta bands of *AF*7 and beta band of

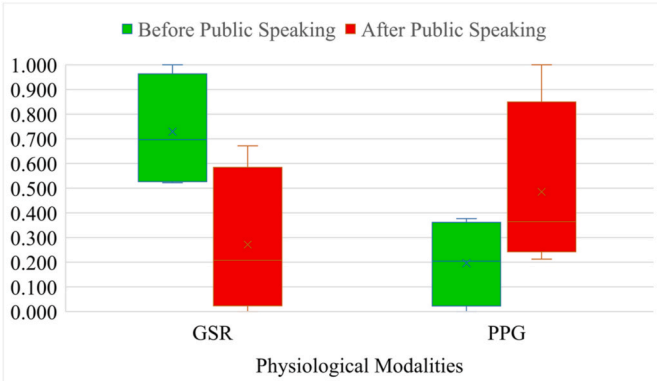

Fig. 4. Boxplot for GSR and PPG signals for participants in rest-state and during public speaking activity phase.

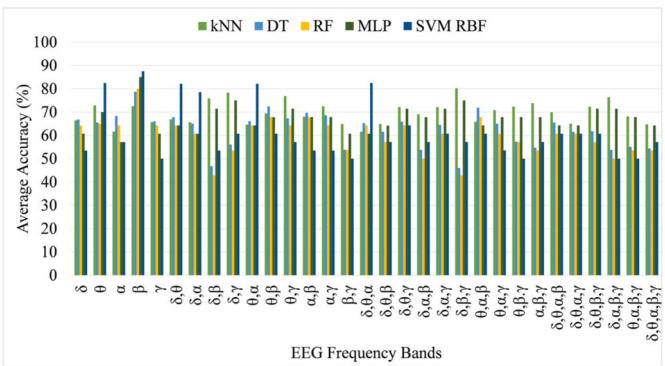

Fig. 5. Classification accuracy of various classifiers based on features extracted from different combinations of EEG frequency bands.

*AF*8 have visually distinct brain activity during rest-state and public speaking activity. Moreover, statistical analysis reveals that there exists a statistical difference in the GSR (a p-value of 0.04) and PPG (a p-value of 0.002) data acquired during the rest-state and public speaking activity. Fig. 4 shows the box plot representation of the GSR and PPG data acquired during rest-state and public speaking activity. It is visually evident from the box plot that the GSR and PPG data acquired in the two phases of the experiment are different from each other and hence reinforcing the outcome of the statistical *t*-test.

3.2. Stress classification

In this study, we have adopted a leave-one-out cross-validation scheme for the training and testing of our model. The leave-one-out scheme is a type of cross-validation scheme in which the number of the fold used for validation is equal to the number of instances used in the experiment. One of the data instances is left out for the testing purpose and the training of the classifier is performed on all the rest of the instances. This process is repeated by treating each instance of the data as a test sample one by one and the achieved accuracy for all the repetitions is averaged out to achieve the final accuracy. Evaluation parameters adopted for the proposed scheme include classification accuracy, F-measure, and kappa statistics. The accuracy of a classifier depicts the percentage of the correctly identified instances from among all the instances. F-measure is a measure of the accuracy of the test and is written as a weighted harmonic mean of precision and recall of the classifier. Kappa statistics reports the amount of inter-rater reliability for categorical data. A value of less than 0 means no agreement, whereas a value in the range of $0.81 - 1.0$ is termed as perfect agreement.

3.2.1. Performance analysis of band and feature selection and fusion

We intended to identify the frequency band, which provides the best discrimination among the EEG data acquired during the rest-state and public speaking activity. Therefore, the acquired EEG data is applied upon with the band selection algorithm [8] to identify the most prominent EEG frequency band contributing to the correct classification of human stress. In the band selection algorithm, a leave-one-out cross--validation scheme is used in which the number of folds is equal to the number of instances in the experiment. The training of the data is performed on all the instances except one, and this procedure is repeated for all the instances. Therefore, average accuracy is computed for all the folds to select an optimum EEG frequency band for feature extraction. Fig. 5 shows the bar-graph of average accuracy obtained by applying the band selection algorithm on the recorded EEG signals. It is evident from the graph that the beta band results in the highest classification accuracy out of all the frequency band combinations.

The extracted features from the beta EEG band are subjected to the recursive feature elimination technique to select the most appropriate feature subset. The selected subset of features consist of MP (β_{AF7}, β_{AF8}, β_{TP9}, AI ($\beta_{AF7,AF8}$), and RASM ($\beta_{AF7,AF8}$). Applying recursive feature elimination on the GSR data results in two selected features i.e., variance and entropy, whereas for PPG data HR, SDNN, and RMSSD features are selected. Feature vector length obtained from EEG, GSR, and PPG signals constitute 5, 2, and 3 features, respectively. These selected features are merged to form a feature vector, which is subjected to different classification algorithms to classify human stress. The selected subset of fea-

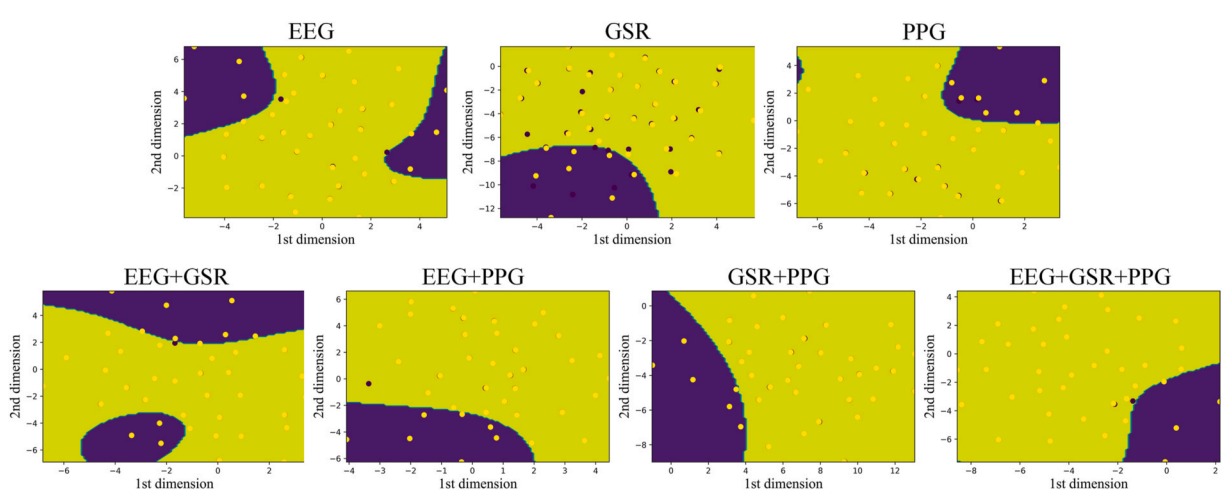

Fig. 6. Selected features obtained from different modalities represented in two dimensions for human stress classification in response to public speaking.

Table 1
Performance comparison of the proposed stress classification framework in terms of average accuracy (Acc (%)), F-measure (F_m), and Kappa values (K) using selected features from individual modality and fusion of selected features from all combinations of EEG, GSR, and PPG signals.

Classifier	EEG			GSR			PPG			EEG + GSR			EEG + PPG			GSR + PPG			EEG + GSR + PPG		
	Acc	F_m	K	Acc	F_m	K	Acc	F_m	K	Acc	F_m	K	Acc	F_m	K	Acc	F_m	K	Acc	F_m	K
kNN	72.50	0.67	0.36	67.50	0.64	0.29	68.75	0.67	0.32	78.75	0.77	0.35	81.25	0.80	0.63	80.00	0.79	0.61	83.75	0.83	0.71
DT	78.75	0.78	0.75	75.00	0.72	0.44	76.25	0.76	0.42	82.50	0.81	0.55	83.75	0.81	0.65	81.25	0.80	0.61	86.25	0.86	0.73
RF	80.00	0.78	0.52	72.50	0.68	0.35	75.00	0.72	0.44	81.25	0.80	9.53	82.50	0.81	0.63	78.75	0.76	0.60	85.00	0.85	0.72
MLP	85.00	0.83	0.64	77.50	0.73	0.46	80.00	0.80	0.61	83.75	0.81	0.54	87.50	0.85	0.74	78.75	0.77	0.61	93.75	0.93	0.83
RBF SVM	**87.50**	**0.86**	**0.76**	**78.75**	**0.74**	**0.47**	**80.00**	**0.79**	**0.72**	**85.00**	**0.78**	**0.57**	**88.75**	**0.86**	**0.78**	**86.25**	**0.85**	**0.65**	**96.25**	**0.96**	**0.87**

Table 2
Cohen's kappa symmetric measure for RBF-SVM classifier using fusion of EEG, GSR and PPG selected features.

	Value	Asymp. Std Error	Approx. T^b	Approx. Sig.
Measure of Kappa agreement	0.875	0.011	14.432	0.0007
No of Valid Cases	80	–	–	–

Table 3
Confusion matrices for the two class stress classification using various machine learning classifier.

S	NS	Classified as	Precision	Recall
33	7	NS=Non-stressed	0.84	0.82
6	34	S=Stressed	0.82	0.85
(a) kNN				
S	NS	Classified as	Precision	Recall
35	5	NS=Non-stressed	0.85	0.87
6	34	S=Stressed	0.87	0.85
(b) DT				
S	NS	Classified as	Precision	Recall
32	8	NS=Non-stressed	0.88	0.80
4	36	S=Stressed	0.81	0.90
(c) RF				
S	NS	Classified as	Precision	Recall
36	4	NS=Non-stressed	0.97	0.90
1	39	S=Stressed	0.90	0.97
(d) MLP				
S	NS	Classified as	Precision	Recall
39	1	NS=Non-stressed	0.95	0.97
2	38	S=Stressed	0.97	0.95
(e) SVM				

tures is visualized in two dimensions using t-Distributed Stochastic Neighbor Embedding (t-SNE) scheme [69]. The t-SNE is a dimensionality reduction technique and is commonly employed to visualize data in 2D or 3D. Moreover, decision boundaries are drawn by Voronoi-based visualization technique [70], where the given set of points are partitioned into multiple regions to create a boundary for different classes. Fig. 6 shows the t-SNE plots along with the decision boundaries for the selected set of features from individual modality as well as the fusion of all possible combinations from three modalities. The fusion of selected features from all three modalities results in a clear and crisp boundary between the stressed and non-stressed groups.

Performance of the proposed stress classification framework in terms of classification accuracy, F-measure, and kappa values by using selected features from the individual modality and all combinations of fusion of EEG, GSR, and PPG selected features with a variety of machine learning algorithms is presented in Table 1. The results depict the fact that for all the classifiers, the fusion of selected features from all three modalities resulted in the highest classification accuracy when compared to individual and combination of two modalities. Moreover, among all the classification accuracies obtained by using the fusion of features from all modalities, the highest accuracy of 96.25% is obtained by using an SVM classifier with RBF kernel. The results depict the fact that along with the highest classification accuracy, an F-measure of 0.962, and a kappa statistics value of 0.875 is attained by the SVM RBF classifier. The kappa symmetric measures for the best case i.e., the fusion of selected features from all three modalities using SVM classifier with RBF kernel are shown in Table 2. A kappa agreement value of 0.875 and a p-value of 0.0007 shows that the agreement obtained by the kappa statistics is not only strong by chance rather it is statistically significant with a p-value quite low than the significance level of 0.05.

Confusion matrices obtained by fusing the selected features from all three modalities using five different classifiers are shown in Table 3. A

Table 4
Performance Comparison of the proposed framework with the state-of-the-art methods for multimodal human stress classification.

Method, Year	Number of Participants	Stressor	Modalities	Classifier	Accuracy (Classes)
[46], 2013	10	Public Speaking	HRV	DT	76.40% (2)
[47], 2015	5	Public Speaking, Cognitive Task	GSR, PPG	SVM	78.97% (2)
[48], 2017	18	Public Speaking, Cognitive Task	Physiological (GSR, PPG, HRV), Sociometric (microphone, accelerometer, proximity, IR)	AdaBoost	79.00% (2) 89.00% (2)
			Physiological + Sociometric Sensors		94.00% (2)
Proposed	**40**	**Public Speaking**	**PPG**	**SVM-RBF**	**80.00% (2)**
			GSR, PPG		**86.25% (2)**
			EEG, GSR, PPG		**96.25% (2)**

confusion matrix is commonly used to explain the performance of the classification algorithm. It shows the number of both correct and incorrect instances for each class based on the true labels of the test data. A confusion matrix is composed of four different components which include true positives, true negatives, false positives, and false negatives. True positives are the entities in which a particular condition occurred, and the classification algorithm also predicted its occurrence. True negative refers to the units in which the true and the predicted label both depict that the event did not occur. Moreover, false-positive identifies the number of instances in which the particular event did not occur but the classification algorithm predicted it to have occurred whereas, false-negative highlights the number of instances in which an event has occurred but the classifier did not predict its occurrence. The confusion matrices of the kNN, DT, RF, MLP, and SVM classifiers illustrates that 33, 35, 32, 36, and 39 instances out of 40 are correctly classified for non-stressed class respectively. Similarly, 34, 34, 36, 39, and 38 instances out of 40 are correctly classified for the stressed class using kNN, DT, RF, MLP, and SVM classifiers respectively. Moreover, the precision and recall obtained for the SVM RBF classifier are higher as compared to other classification algorithms used in the study.

3.2.2. *Performance comparison and discussion*

The comparison of the proposed scheme with the existing state-of-the-art acute stress classification framework in response to public speaking activity is presented in Table 4. The comparison of the developed stress classification scheme with these methods is performed based on the number of participants of the experiment, stressor adopted, modalities, classifier used, and the classification accuracy achieved along with the number of classes. A comparison performed by observing only the accuracy achieved by the fusion of all modalities is not fair as they have used different physiological sensors. Therefore, we have also compared the accuracies achieved by the modalities that are common between our method and methods available in the literature. HRV data has been used for human stress classification in response to the public speaking task and achieved an accuracy of 76.40% for two states using a decision tree classifier [46]. Our method when only used HRV based features from PPG signals has achieved an accuracy of 80.0%. A human stress assessment scheme using a fusion of GSR and PPG signals in response to the cognitive and public speaking task is presented in Ref. [47]. SVM classifier is used for classifying the non-stressed and stressed state and average classification accuracy of 78.97% is achieved. The fusion of features from GSR and PPG signals used by our proposed method has achieved an accuracy of 86.25%. In Ref. [48] human stress assessment using physiological signals (GSR, PPG, and HRV), sociometric sensors (microphone, accelerometer, proximity, and infrared sensors), and fusion of physiological signals and sociometric sensors in response to public speaking is presented. The extracted set of features are subjected to a correlation-based feature selection method and the AdaBoost classifier is used for binary stress classification. The accuracy achieved by the physiological signals, sociometric sensors, and fusion of physiological signals and sociometric sensors are 79.0%, 89.0%, and 94.0% respectively. When we compared our method with the physiological signals common with [48] i.e., PPG and GSR have achieved an accuracy of 86.25%.

The number of participants in our proposed scheme is 40, which is highest when compared to the other stress measurement methods available in the literature while using public speaking as a stressor. Our proposed scheme for stress classification outperforms all these methods and achieves an accuracy of 96.25% for two-state stress classification using the fusion of selected features from EEG, GSR, and PPG signals. It is important to note that the proposed framework uses EEG signals with other modalities for the first time to classify human stress in response to a public speaking activity in front of a real audience and achieves the highest classification accuracy.

4. Conclusion and future work

In this study, physiological signals are used for the classification of stress in response to real-life public speaking activity. Wearable sensors are used to acquire the EEG, GSR, and PPG signals during both the rest-state and public speaking activity. Features are extracted from the recorded physiological signals and classification is performed by using selected features from both individual sensors and by fusion of features from all the modalities. Recursive feature elimination technique and SVM classifier with RBF kernel provide the highest classification accuracy of 96.25% via leave-one-out cross-validation scheme using the fusion of selected features. In future studies, the number of participants in the experiment can be increased. EEG headsets with more electrodes can be used to record brain activity from different brain regions to make the results of the proposed scheme more robust and reliable. In the future deep learning models can also be used to enhance classification accuracy.

Declaration of competing interest

None Declared.

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